

FreeRTOS를 이용한 LED Demo 실습

FreeRTOS 다운로드

□ FreeRTOS 검색

× | 🗣️ 🎤 📷

AI 모드 전체 쇼핑 이미지 동영상 뉴스 짧은 동영상 더보기 ▾ 도구 ▾

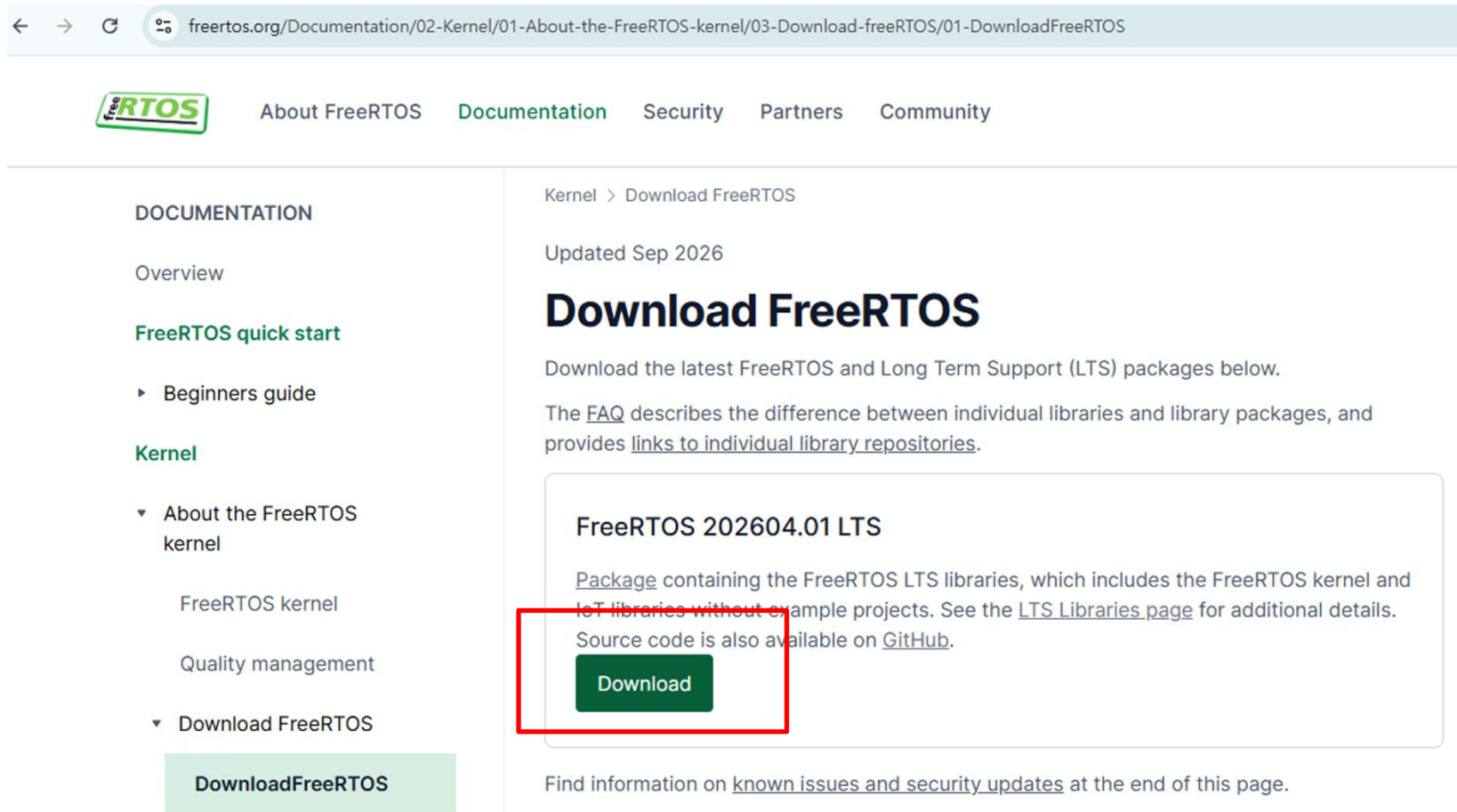
 FreeRTOS
<https://www.freertos.org> > Documentation > 02-Kernel ⋮

Download FreeRTOS

Download the latest FreeRTOS and Long Term Support (LTS) packages below. The FAQ describes the difference between individual libraries and library packages, and ...

FreeRTOS 다운로드

□ FreeRTOS 다운로드



The screenshot shows the FreeRTOS website's download page for the kernel. The browser address bar displays the URL: freertos.org/Documentation/02-Kernel/01-About-the-FreeRTOS-kernel/03-Download-freeRTOS/01-DownloadFreeRTOS. The website's navigation bar includes links for About FreeRTOS, Documentation (highlighted), Security, Partners, and Community. The left sidebar contains a 'DOCUMENTATION' section with links to Overview, FreeRTOS quick start, and Kernel. Under the Kernel section, there are links for 'About the FreeRTOS kernel' and 'Download FreeRTOS'. The 'Download FreeRTOS' link is highlighted with a green background. The main content area shows the breadcrumb 'Kernel > Download FreeRTOS', the update date 'Updated Sep 2026', and the title 'Download FreeRTOS'. Below the title, it states 'Download the latest FreeRTOS and Long Term Support (LTS) packages below.' and provides a link to the FAQ. A highlighted box contains the section 'FreeRTOS 202604.01 LTS', a description of the package, and a green 'Download' button. At the bottom, it mentions finding information on known issues and security updates at the end of the page.

freertos.org/Documentation/02-Kernel/01-About-the-FreeRTOS-kernel/03-Download-freeRTOS/01-DownloadFreeRTOS

About FreeRTOS Documentation Security Partners Community

DOCUMENTATION

Overview

FreeRTOS quick start

▸ Beginners guide

Kernel

▼ About the FreeRTOS kernel

FreeRTOS kernel

Quality management

▼ Download FreeRTOS

DownloadFreeRTOS

Kernel > Download FreeRTOS

Updated Sep 2026

Download FreeRTOS

Download the latest FreeRTOS and Long Term Support (LTS) packages below.

The [FAQ](#) describes the difference between individual libraries and library packages, and provides [links to individual library repositories](#).

FreeRTOS 202604.01 LTS

[Package](#) containing the FreeRTOS LTS libraries, which includes the FreeRTOS kernel and IoT libraries without example projects. See the [LTS Libraries page](#) for additional details. Source code is also available on [GitHub](#).

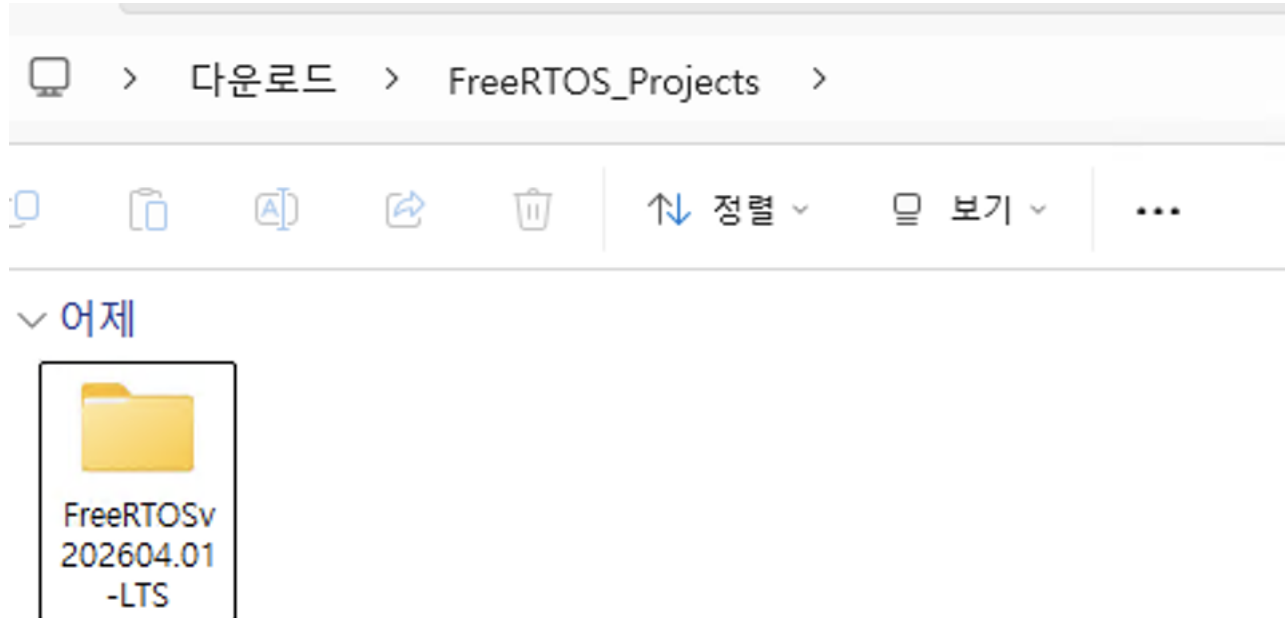
Download

Find information on [known issues and security updates](#) at the end of this page.

○ 다운받은 파일(FreeRTOSv202604.01-LTS.zip) 압축 해제

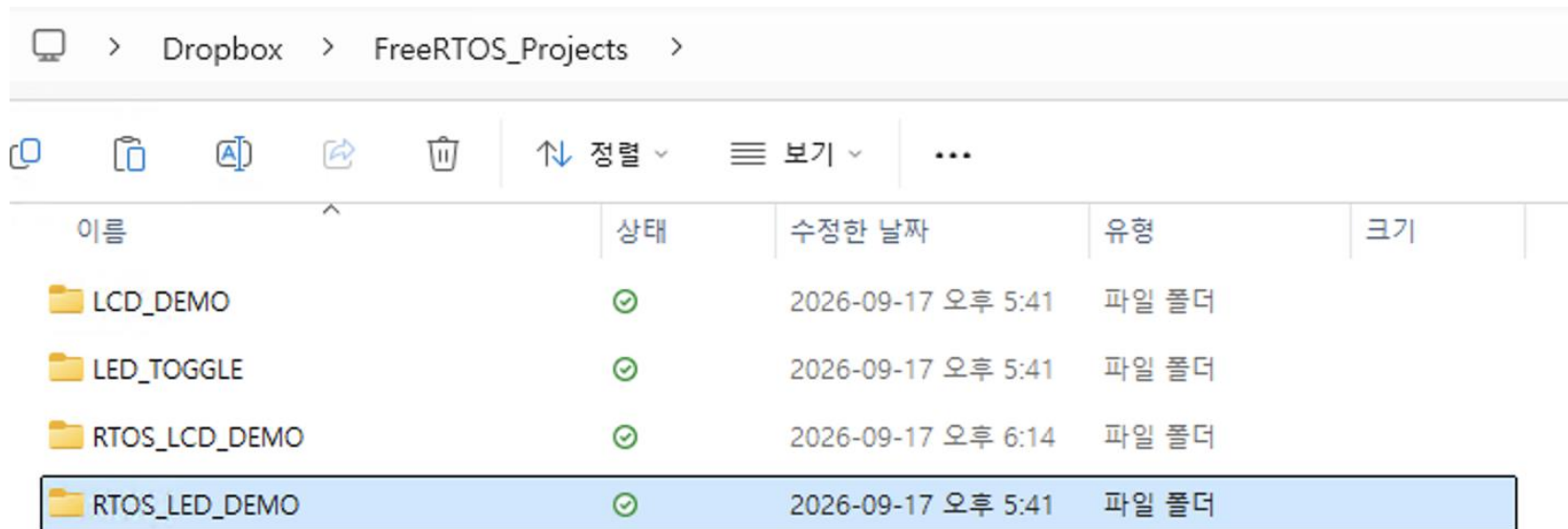
FreeRTOS 다운로드

- FreeRTOS 소스코드를 'FreeRTOS_Projects' 디렉터리로 이동



FreeRTOS LED Demo 실습

- 'FreeRTOS_Projects' 디렉터리 내 'RTOS_LED_DEMO' 디렉터리 생성



The screenshot shows a file explorer window with the path 'Dropbox > FreeRTOS_Projects >'. The toolbar includes icons for copy, paste, rename, share, delete, sort, view, and a menu. The file list is as follows:

이름	상태	수정한 날짜	유형	크기
LCD_DEMO	✓	2026-09-17 오후 5:41	파일 폴더	
LED_TOGGLE	✓	2026-09-17 오후 5:41	파일 폴더	
RTOS_LCD_DEMO	✓	2026-09-17 오후 6:14	파일 폴더	
RTOS_LED_DEMO	✓	2026-09-17 오후 5:41	파일 폴더	

FreeRTOS LED Demo 실습

- 'RTOS_LED_DEMO' 디렉터리 내 'Source' 디렉터리 생성

🖥️ > Dropbox > FreeRTOS_Projects > RTOS_LED_DEMO > RTOS_LED_DEMO >					
📄 📁 📄 📄 🗑️ ⬆️ 정렬 ▾ ≡ 보기 ▾ ...					
이름	상태	수정한 날짜	유형	크기	
📁 RTOS_LED_DEMO	✔️	2026-09-17 오후 5:41	파일 폴더		
📁 Source	✔️	2026-09-17 오후 5:41	파일 폴더		
🔧 LED_DEMO	✔️	2026-09-16 오후 10:36	ATMEL Studio 7.0...	1KB	

FreeRTOS LED Demo 실습

- 'FreeRTOS-Kernel' 디렉터리 내 관련 디렉토리 및 소스파일을 생성한 'Source' 디렉터리에 복사

The screenshot shows a file explorer window with the path: Dropbox > FreeRTOS_Projects > RTOS_LED_DEMO > RTOS_LED_DEMO > Source. The search bar contains 'Source 검색'. The main pane displays a list of files and folders:

이름	상태	수정한 날짜	유형	크기
include	✓	2026-09-17 오후 5:41	파일 폴더	
portable	✓	2026-09-17 오후 5:41	파일 폴더	
croutine	✓	2026-09-15 오후 9:59	C Source File	18KB
event_groups	✓	2026-09-15 오후 9:59	C Source File	36KB
heap_1				
list				
port				
queue				
stream_buffer				
tasks				
timers				

An inset window shows the 'FreeRTOS-Kernel' directory. The path is: FreeRTOSv202604.01-LTS > FreeRTOS-LTS > FreeRTOS > FreeRTOS-Kernel. The main pane displays a list of files and folders:

이름	수정한 날짜	유형	크기
include	2026-09-16 오후 11:10	파일 폴더	
portable	2026-09-16 오후 11:10	파일 폴더	
examples	2026-09-15 오후 9:59	파일 폴더	
croutine	2026-09-15 오후 9:59	C Source File	18KB
event_groups	2026-09-15 오후 9:59	C Source File	36KB
list	2026-09-15 오후 9:59	C Source File	10KB
queue	2026-09-15 오후 9:59	C Source File	128KB
stream_buffer	2026-09-15 오후 9:59	C Source File	75KB
tasks	2026-09-15 오후 9:59	C Source File	357KB
timers	2026-09-15 오후 9:59	C Source File	57KB

FreeRTOS LED Demo 실습

- 'FreeRTOS-KernelFreeRTOS-Kernel\portable\WMemMang' 디렉터리 내 'heap_1.c'를 'Source' 디렉터리에 복사

The screenshot shows two overlapping Windows File Explorer windows. The background window displays the 'Source' directory, and the foreground window displays the 'MemMang' directory.

Background Window (Source Directory):

- Path: > Dropbox > FreeRTOS_Projects > RTOS_LED_DEMO > RTOS_LED_DEMO > Source >
- Search: Source 검색
- Files:

이름	상태	수정한 날짜	유형	크기
include	✓	2026-09-17 오후 5:41	파일 폴더	
portable	✓	2026-09-17 오후 5:41	파일 폴더	
croutine	✓	2026-09-15 오후 9:59	C Source File	18KB
event_groups	✓	2026-09-15 오후 9:59	C Source File	36KB
heap_1				
list				
port				
queue				
stream_buffer				
tasks				
timers				

Foreground Window (MemMang Directory):

- Path: > ... FreeRTOS-LTS > FreeRTOS > FreeRTOS-Kernel > portable > MemMang
- Files:

이름	수정한 날짜	유형	크기
heap_1	2026-09-15 오후 9:59	C Source File	7KB
heap_2	2026-09-15 오후 9:59	C Source File	17KB
heap_3	2026-09-15 오후 9:59	C Source File	4KB
heap_4	2026-09-15 오후 9:59	C Source File	24KB
heap_5	2026-09-15 오후 9:59	C Source File	29KB
ReadMe	2026-09-15 오후 9:59	인터넷 바로 가기	1KB

FreeRTOS LED Demo 실습

- ‘portable’ 디렉터리 내 ‘GCC’ 이외 모든 자료 삭제

🖥️ > ... RTOS_LED_DEMO > RTOS_LED_DEMO > Source > portable >				
📄 📁 📄 📄 🗑️ 정렬 보기 ...				
이름	상태	수정한 날짜	유형	크기
📁 GCC	✅	2026-09-17 오후 5:41	파일 폴더	

- ‘portable\GCC’ 디렉터리 내 ‘ATMega323’ 이외 모든 자료 삭제

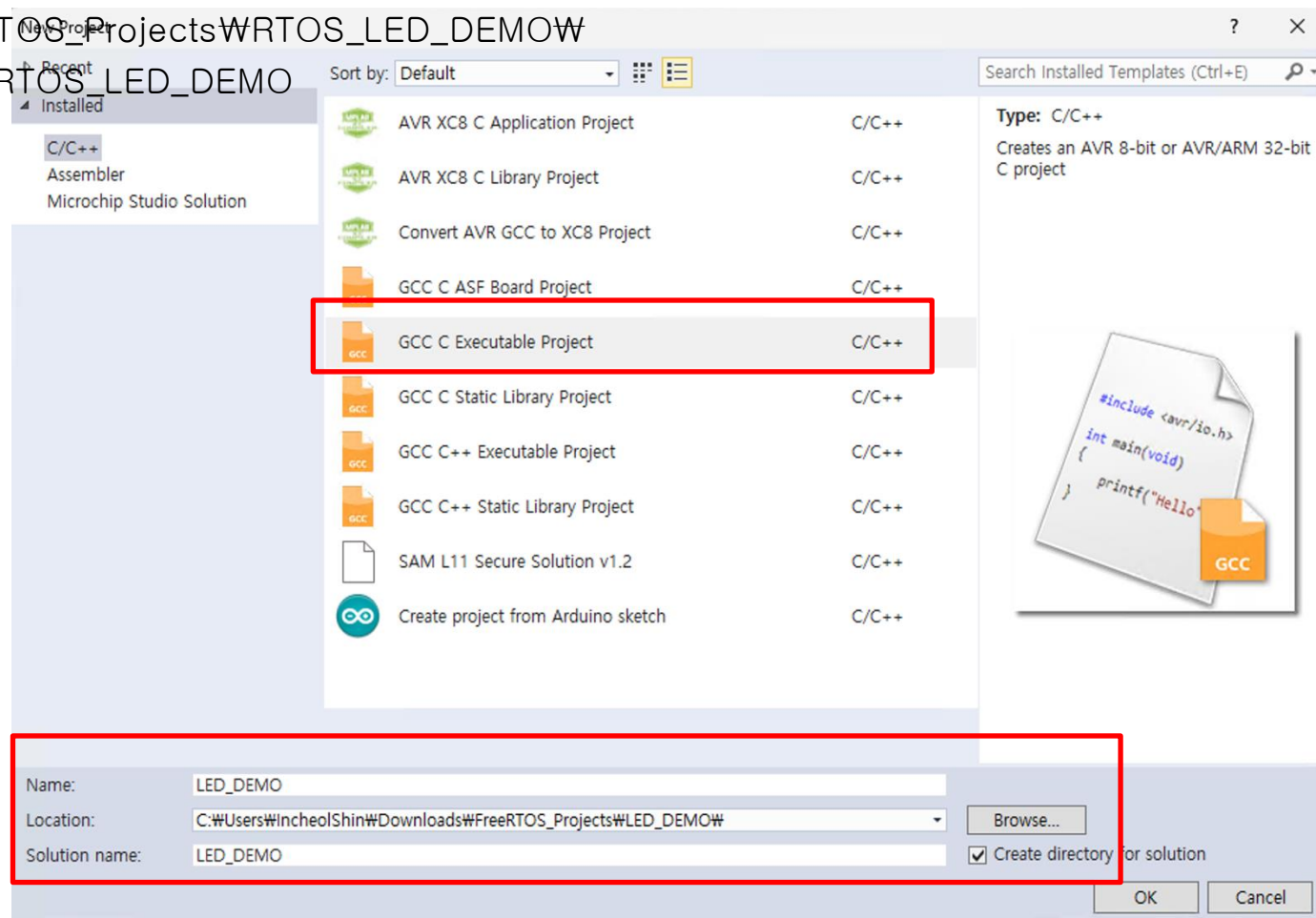
🖥️ > ... RTOS_LED_DEMO > RTOS_LED_DEMO > Source > portable > GCC >				
📄 📁 📄 📄 🗑️ 정렬 보기 ...				
이름	상태	수정한 날짜	유형	크기
📁 ATMega323	✅	2026-09-17 오후 5:41	파일 폴더	

FreeRTOS LED Demo 실습

□ RTOS_LED_DEMO 프로젝트 생성

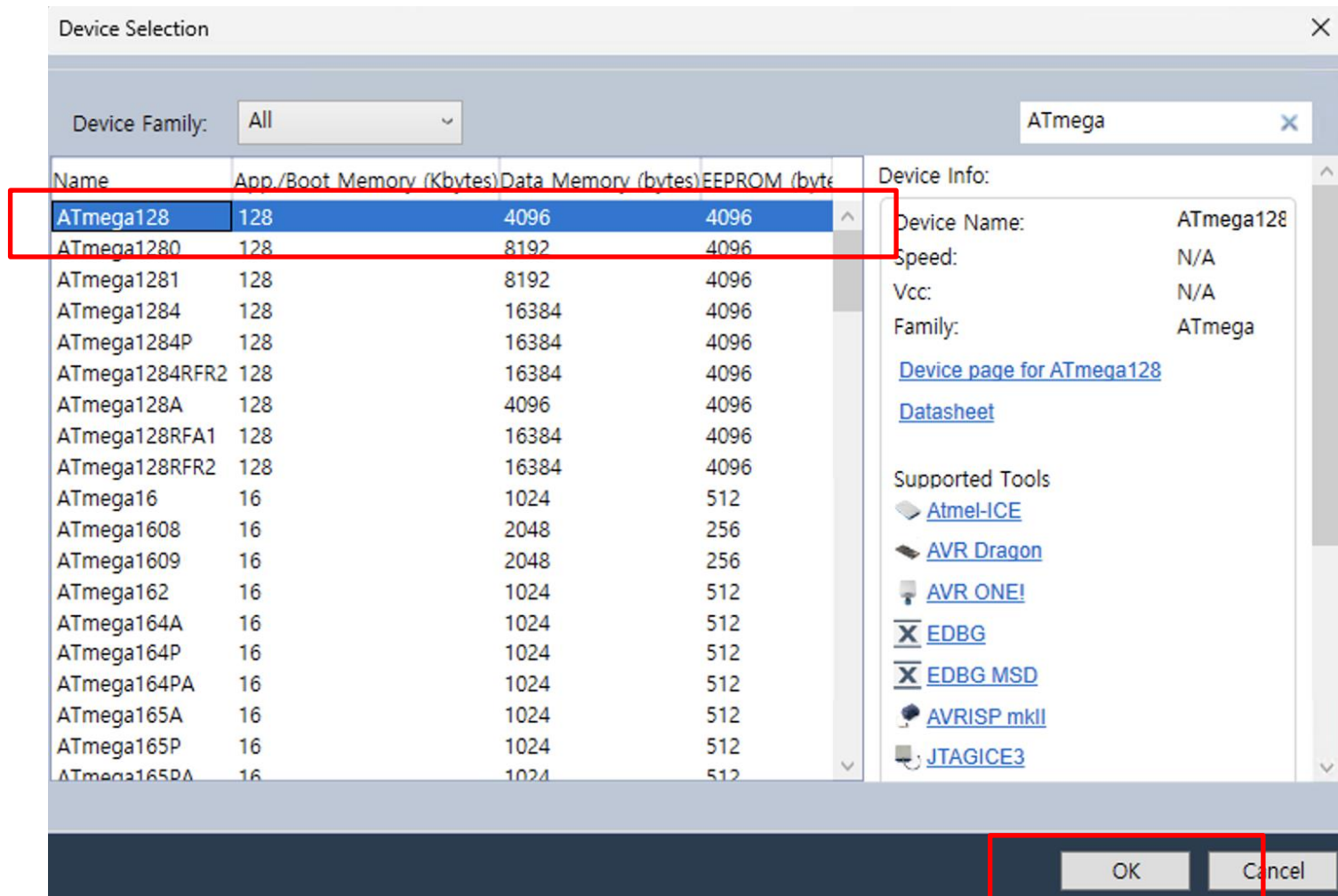
○ Microchip Studio 프로그램 실행 후

- GCC C Executable Project 선택
- Name : RTOS_LED_DEMO
- Location: FreeRTOS_Projects\RTOS_LED_DEMO\
- Solution name: RTOS_LED_DEMO



FreeRTOS LED Demo 실습

- RTOS_LED_DEMO 프로젝트 생성 (계속)
- Microchip Studio 프로그램 실행 후 (계속)
 - ATmega128 선택 후 'OK'



FreeRTOS LED Demo 실습

LED_DEMO - Microchip Studio

Advanced Mode Quick Launch (Ctrl+Q)

File Edit View VAssistX ASF Project Build Debug Tools Window Help

main.c

```
/*
 * LED_DEMO.c
 *
 * Created: 2026-09-16 오후 10:36:15
 * Author : IncheolShin
 */

#define F_CPU 14745600UL
#include <avr/io.h>
#include "FreeRTOS.h"
#include "task.h"

// LED를 제어하는 간단한 태스크
void vLEDBlinkTask(void *pvParameters) {
    // PORTB 전체를 출력으로 설정
    DDRB = 0xFF;
    PORTB = 0x00;
    for( ;; ) {
        PORTB ^= 0xFF; // 모든 LED 상태 반전

        // configTICK_RATE_HZ가 1000이므로 1000 ticks = 1초입니다.
        vTaskDelay(pdMS_TO_TICKS(500));
    }
}

int main(void) {
    // 1. 태스크 생성 (최소 스택 크기 사용, 우선순위 1)
    xTaskCreate(vLEDBlinkTask, "LED_TSK", configMINIMAL_STACK_SIZE, NULL, 1, NULL);

    // 2. 스케줄러 시작 (이 함수가 호출되면 태스크 구동이 시작됩니다)
    vTaskStartScheduler();

    // 메모리 부족 등의 이유로 스케줄러 실행에 실패하면 아래로 내려옵니다.
    while (1) {
        // 에러 표시용 루프
    }
}
```

Solution Explorer

Search Solution Explorer (Ctrl+Q)

Solution 'LED_DEMO' (1 project)

- LED_DEMO
 - Dependencies
 - Output Files
 - Libraries
 - FreeRTOSConfig.h
 - heap_1.c
 - list.c
 - main.c
 - port.c
 - queue.c
 - tasks.c

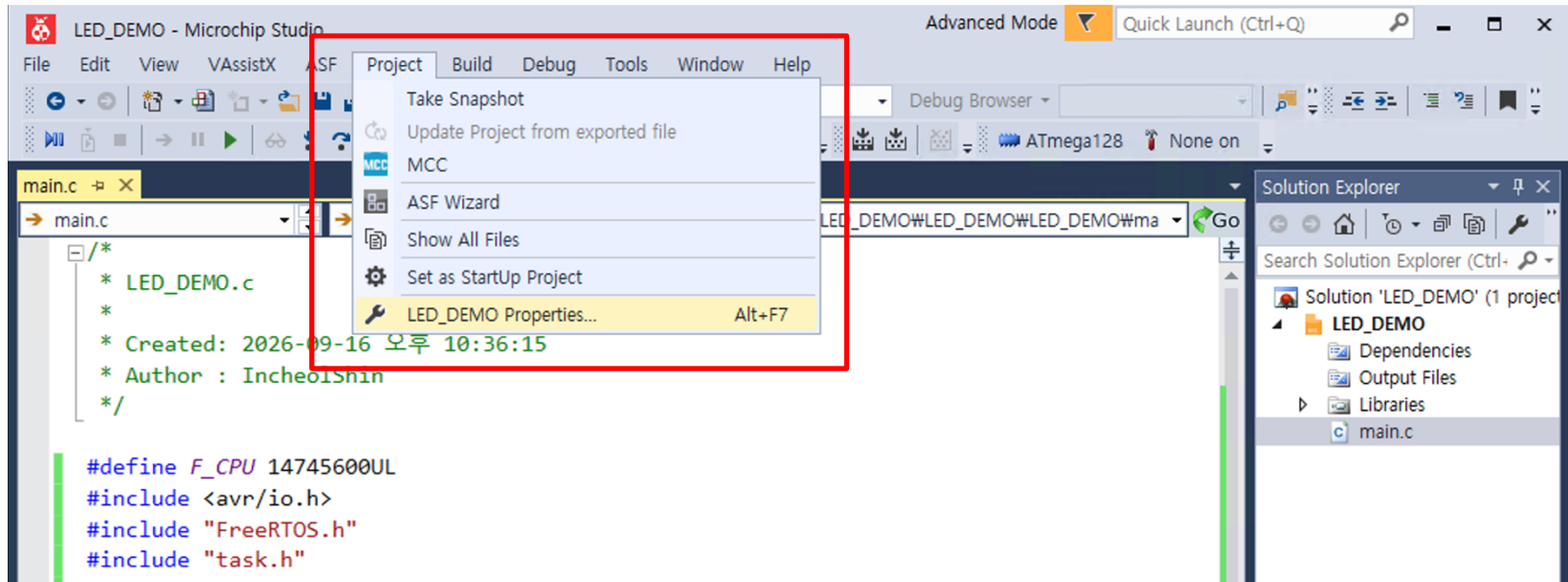
VA View VA Outline Solution E...

Properties

100 %

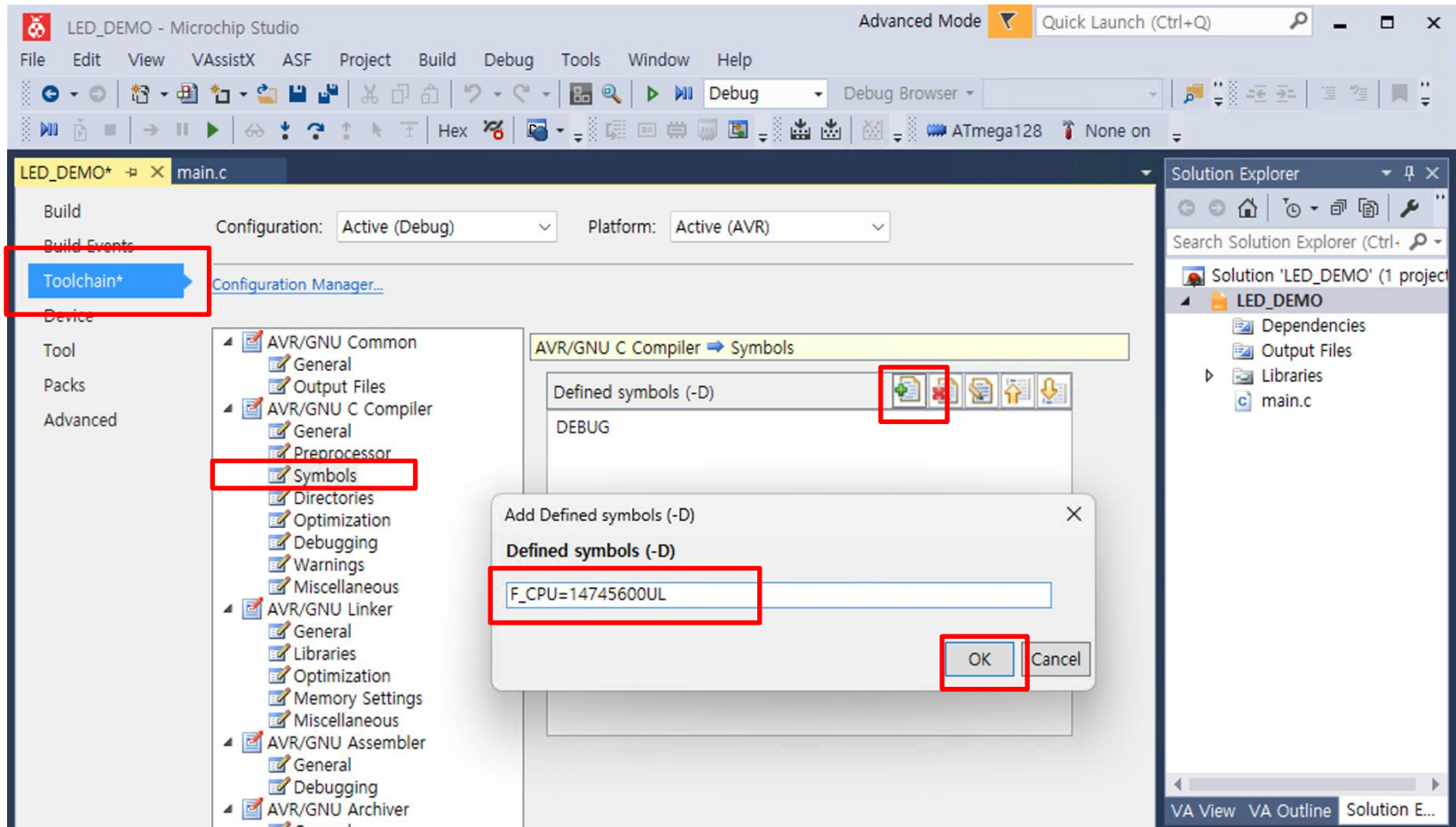
FreeRTOS LED Demo 실습

- 컴파일러 심볼 등록
- 프로젝트 속성 설정 선택



FreeRTOS LED Demo 실습

- 컴파일러 심볼 등록 (계속)
 - 심볼(F_CPU=14745600UL) 등록

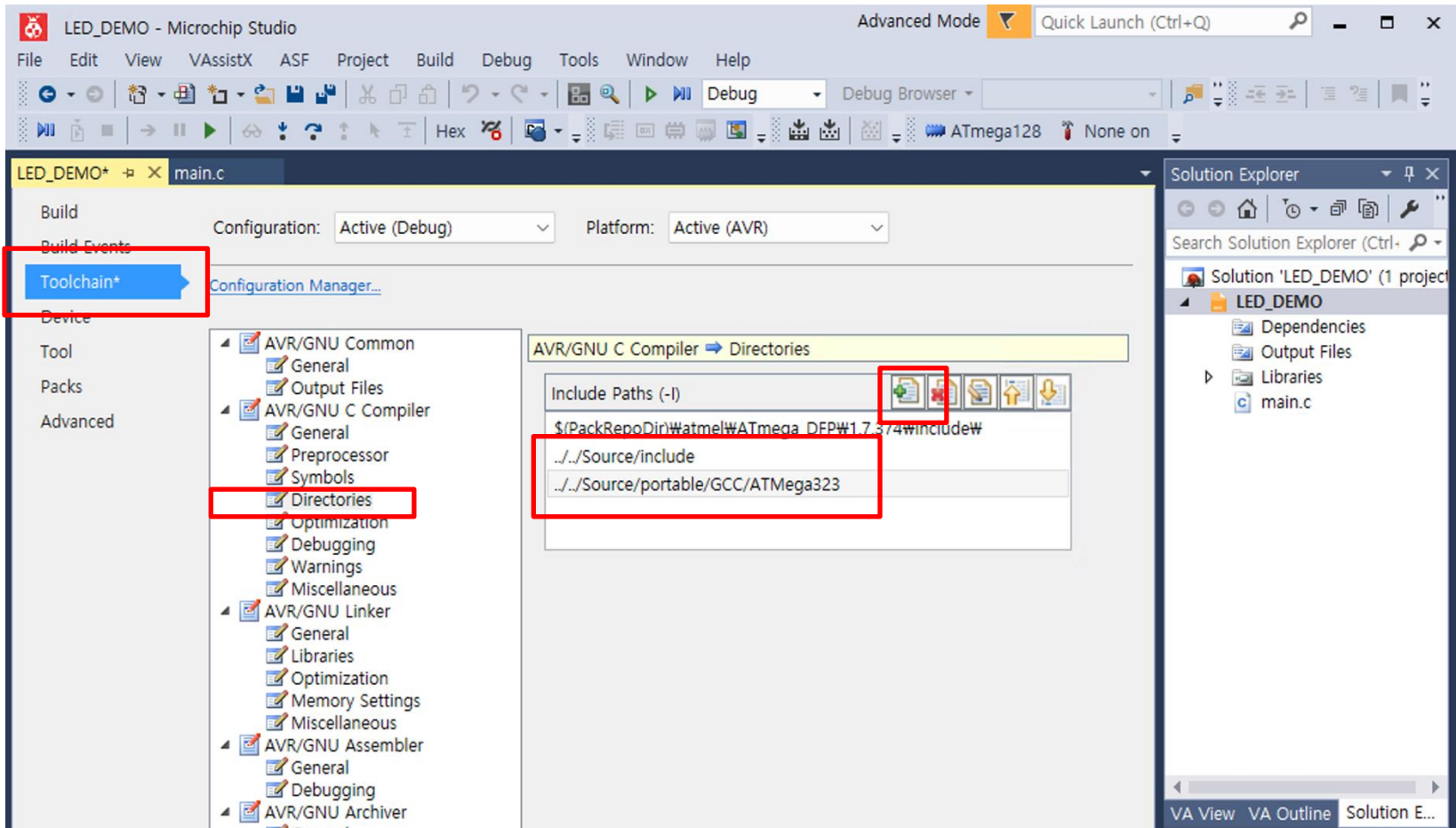


➤ 반드시 저장(ctrl+s)

FreeRTOS LED Demo 실습

□ 컴파일러 디렉터리 등록

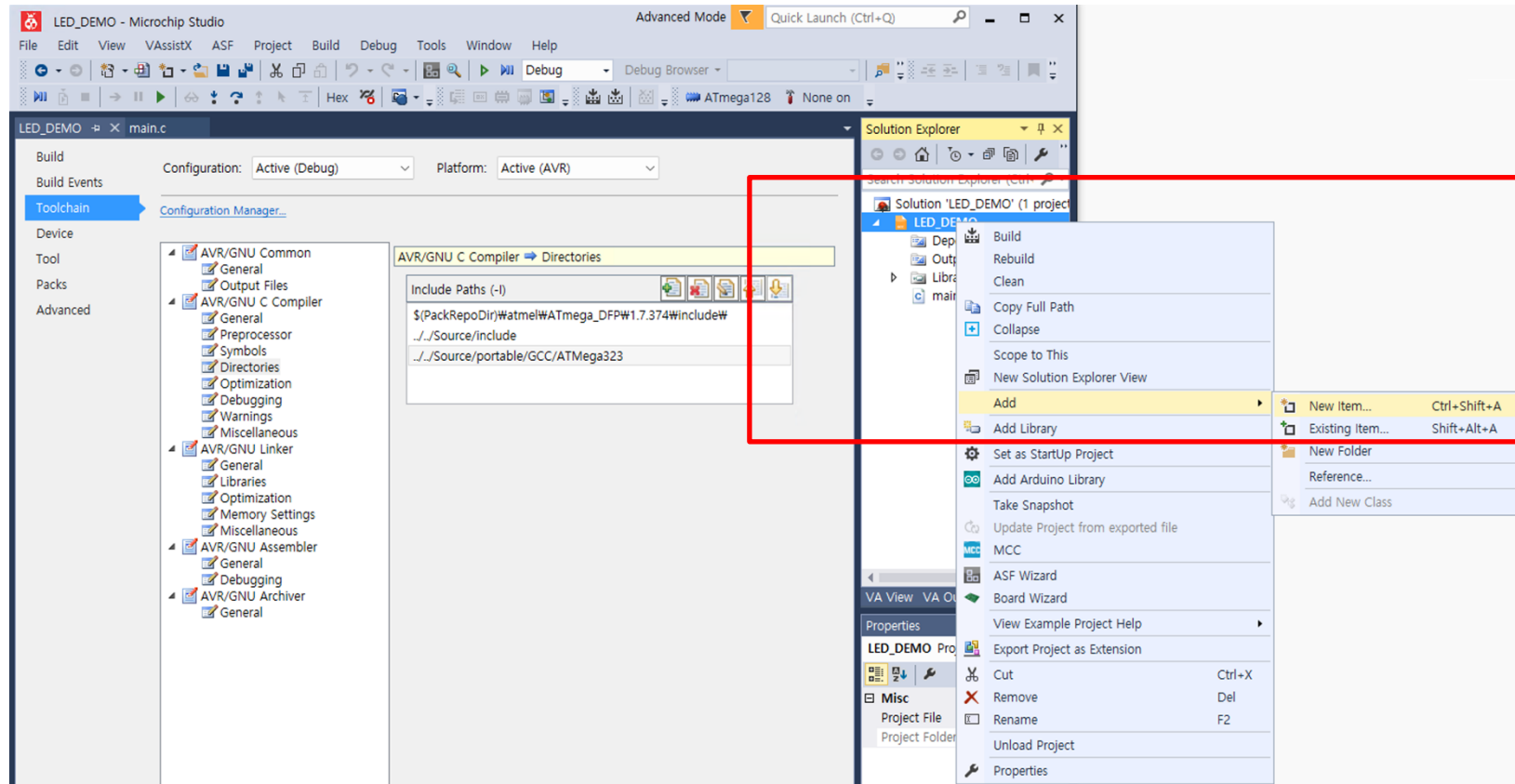
○ ‘...WSourceWinclude’와 ‘...WSourceWportableWGCCWATMega323’ 등록



➤ 반드시 저장(ctrl+s)

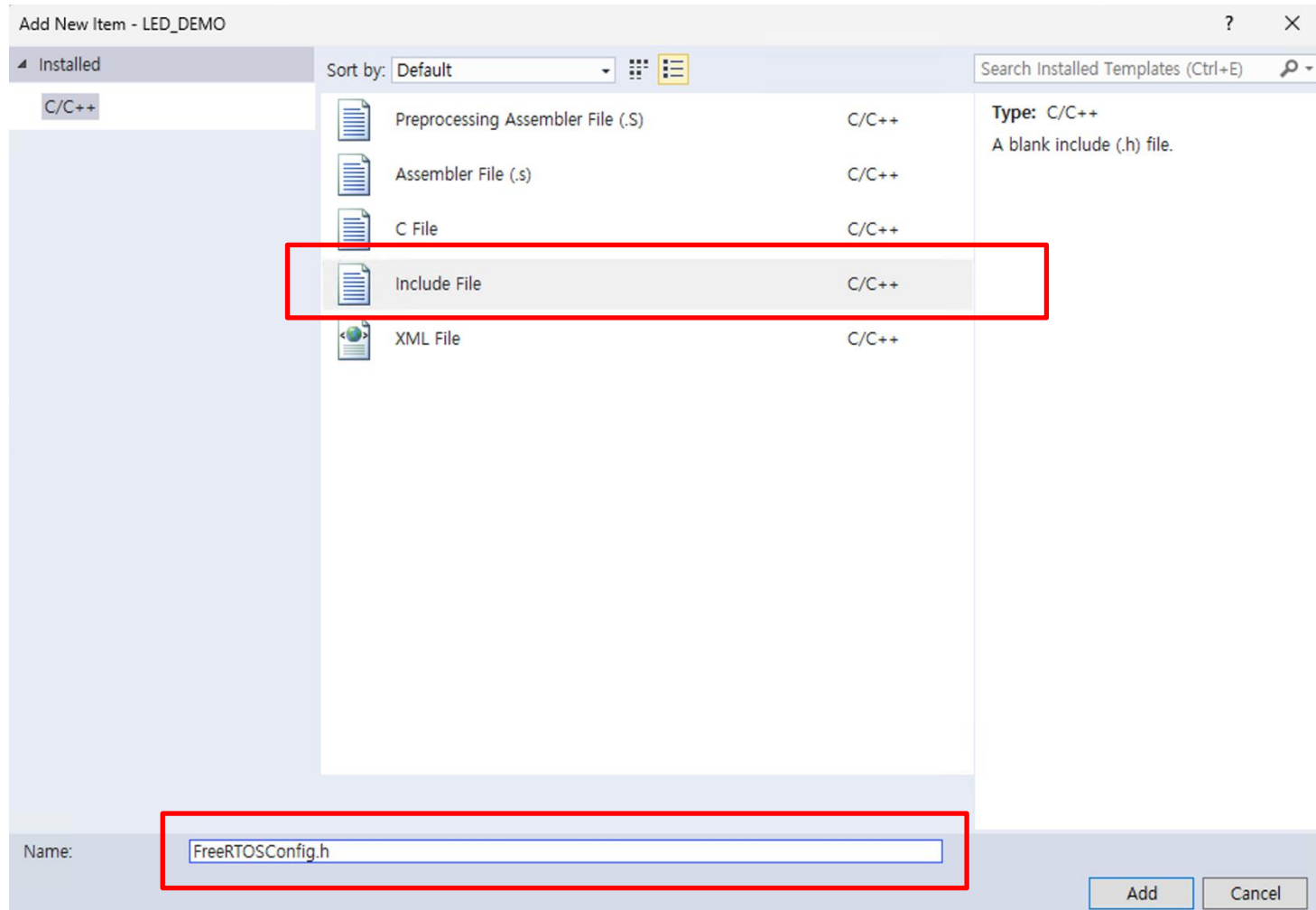
FreeRTOS LED Demo 실습

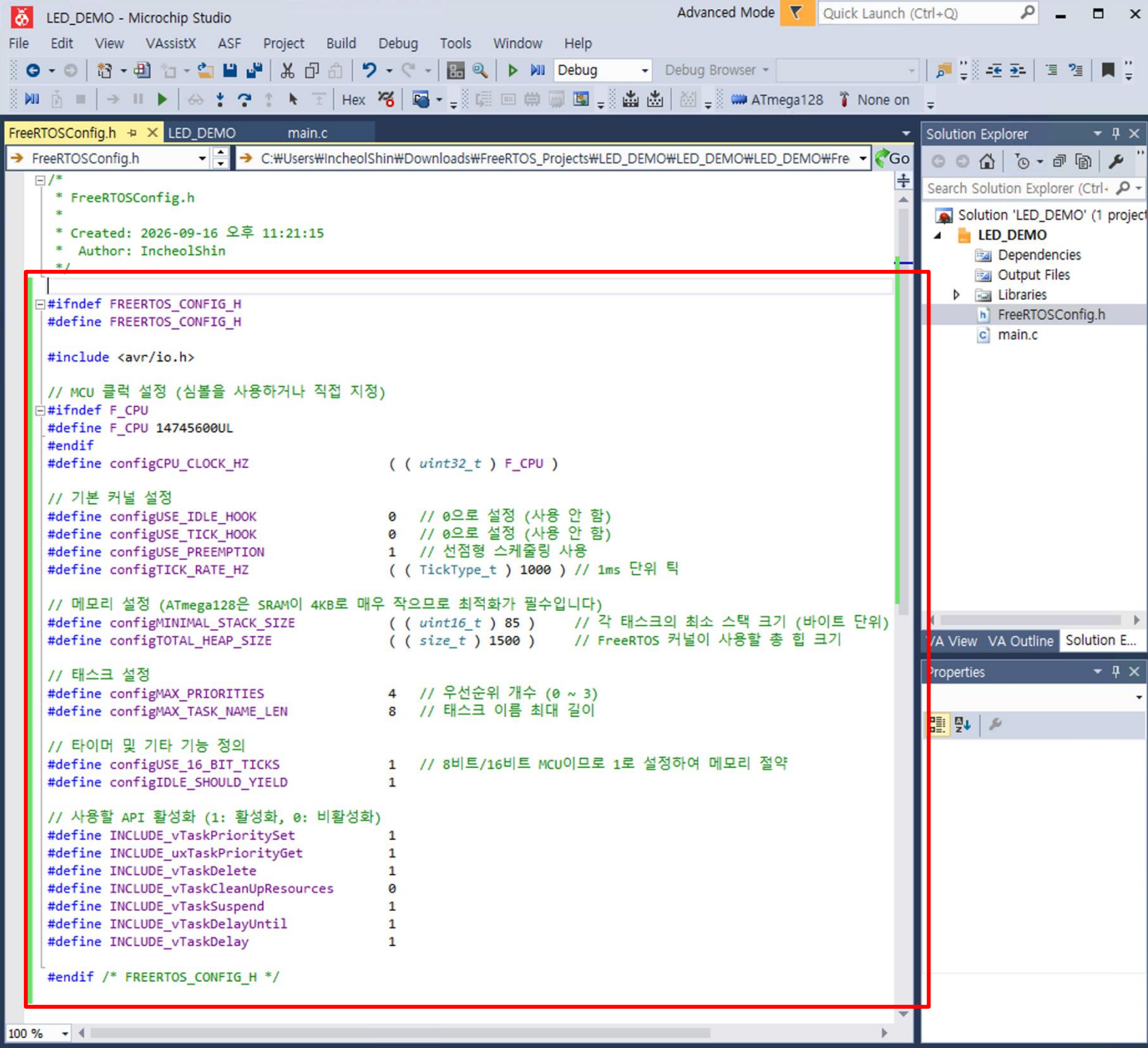
- 'FreeRTOSConfig.h' 헤더파일 작성
- 프로젝트에 'FreeRTOSConfig.h' 헤더파일 추가



FreeRTOS LED Demo 실습

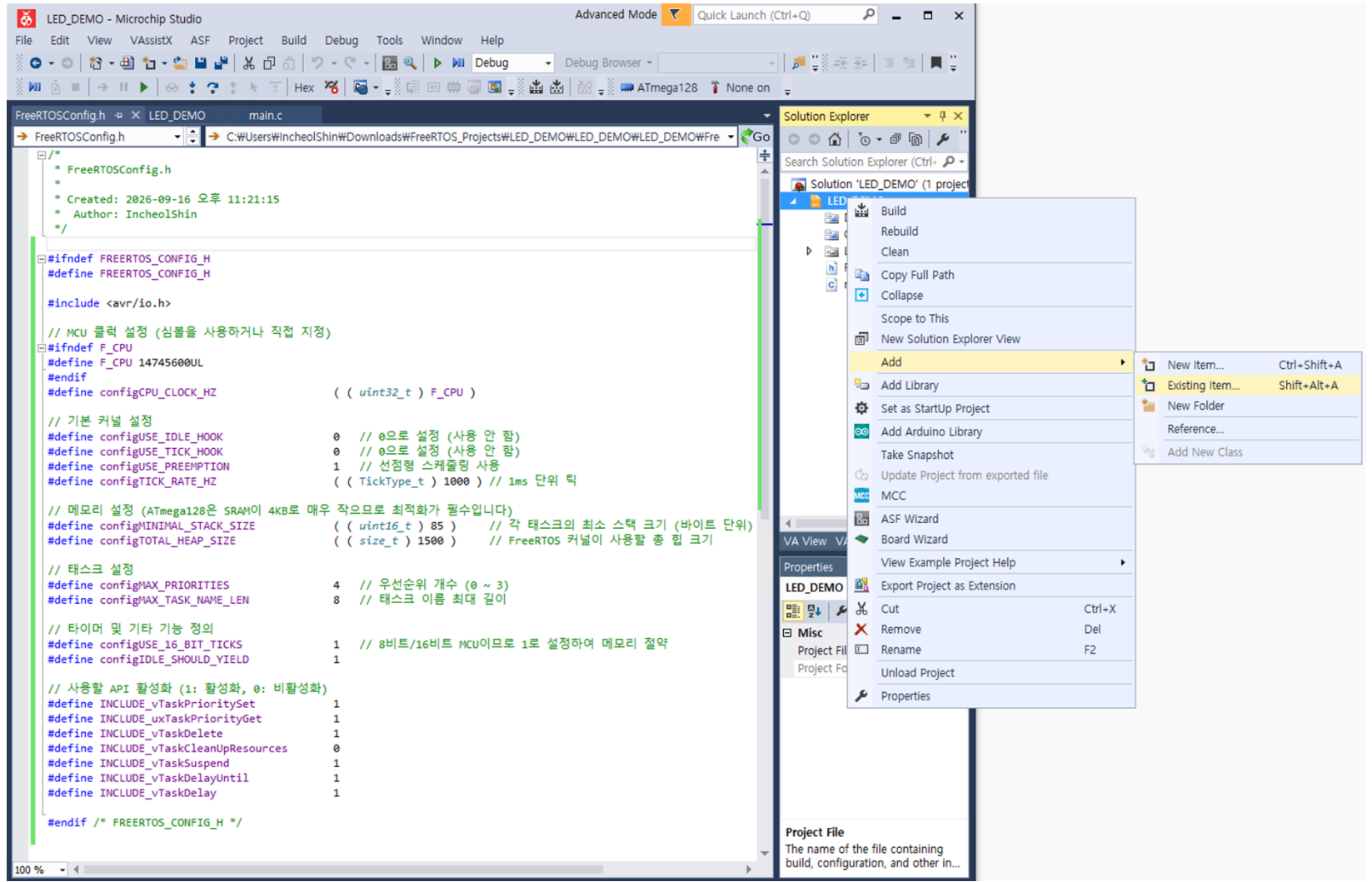
- 'FreeRTOSConfig.h' 헤더파일 작성 (계속)
- 프로젝트에 'FreeRTOSConfig.h' 헤더파일 추가 (계속)





FreeRTOS LED Demo 실습

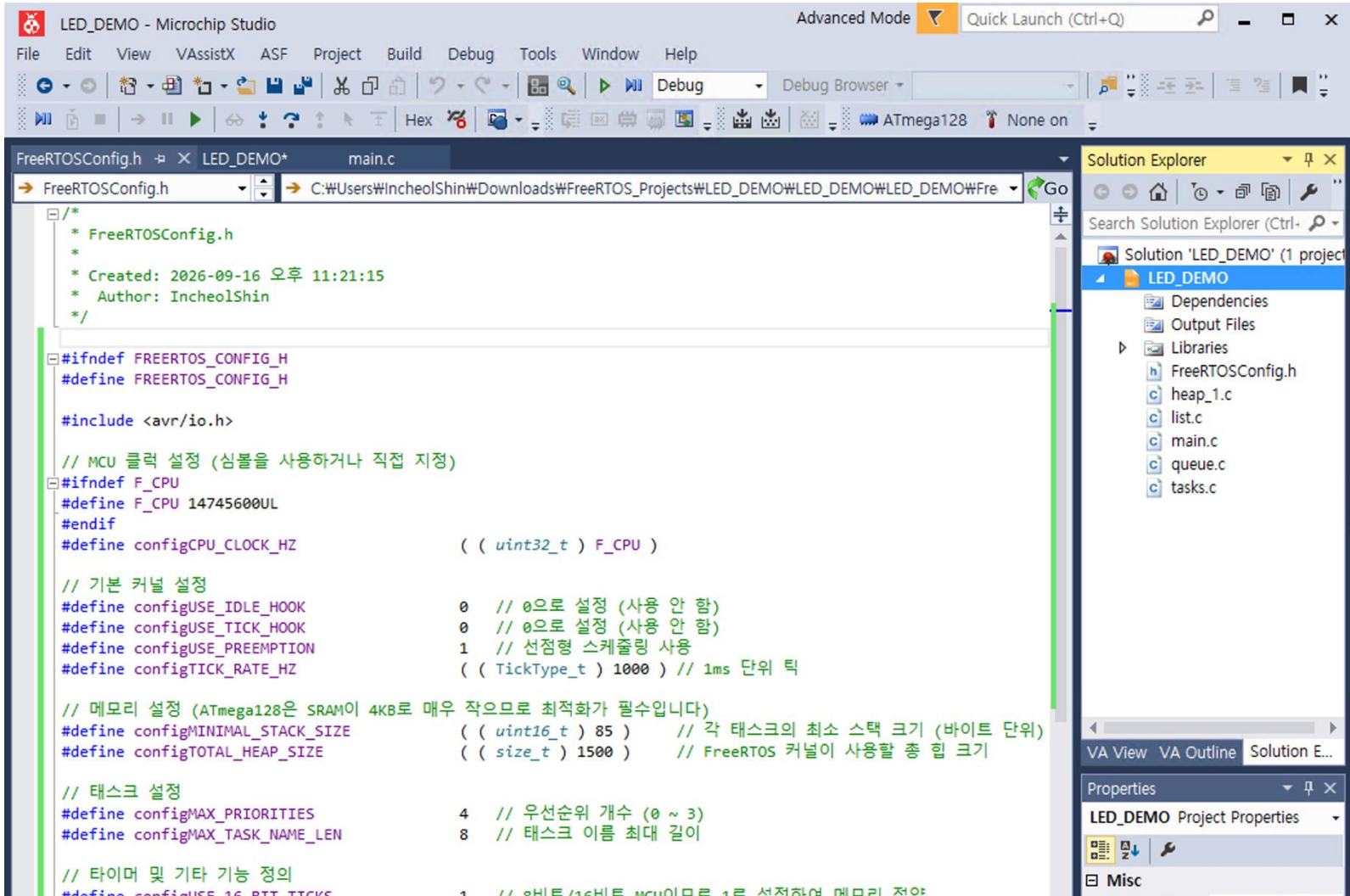
- 기타 소스코드 추가
- 'Existing item' 선택



FreeRTOS LED Demo 실습

□ 기타 소스코드 추가 (계속)

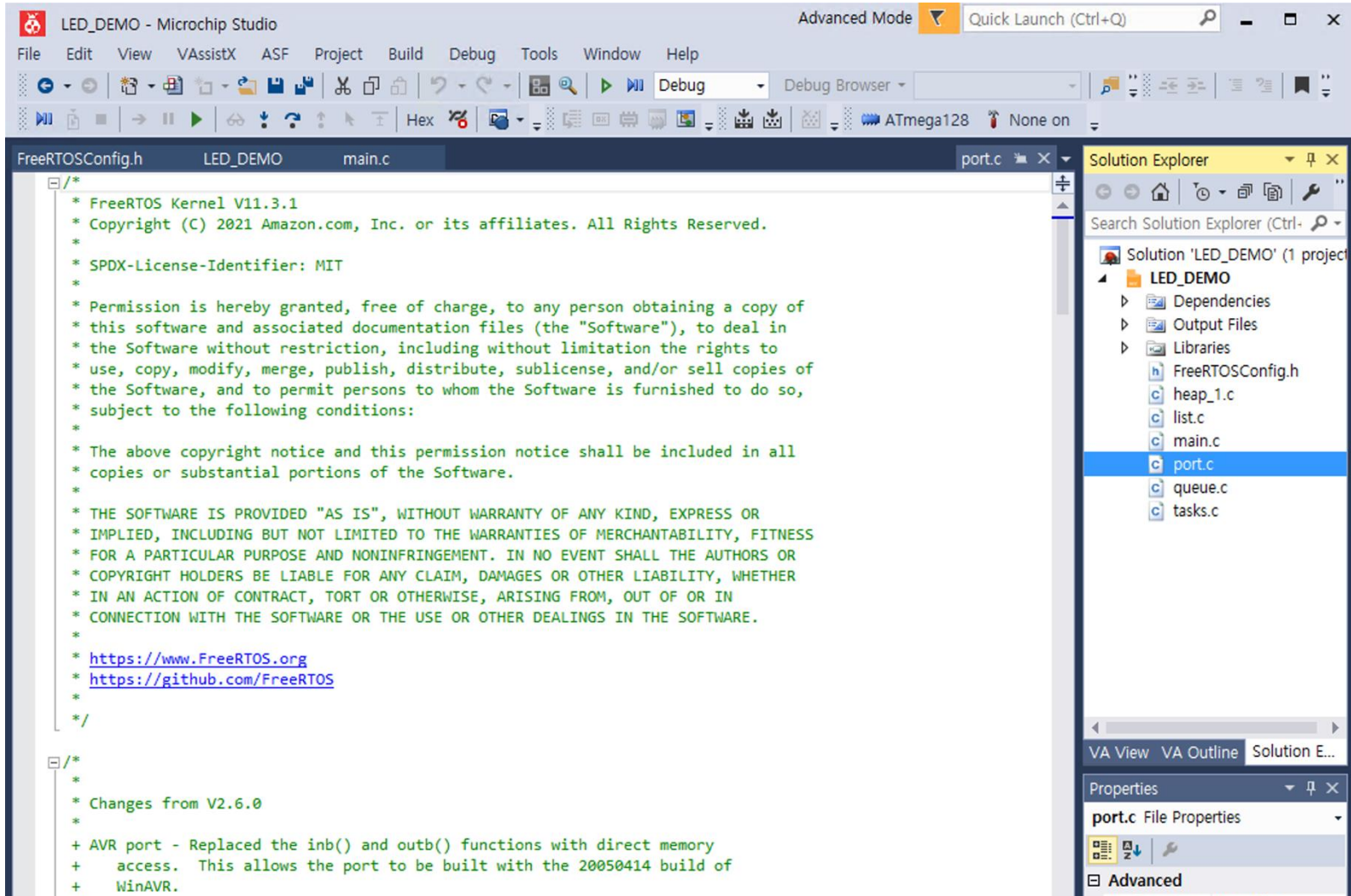
○ 'Source' 디렉터리에 있는 'heap_1.c', 'list.c', 'queue.c', 'tasks.c' 추가



FreeRTOS LED Demo 실습

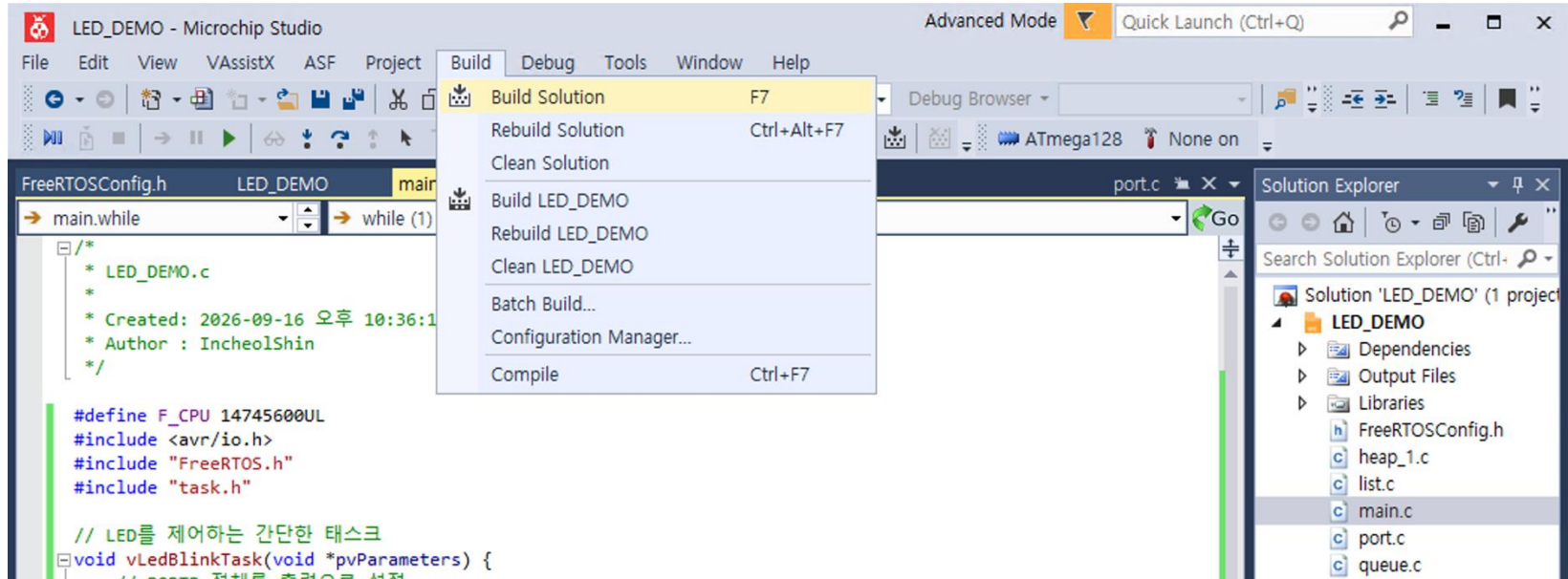
□ 기타 소스코드 추가 (계속)

○ 'LED_DEMO\Source\portable\GCC\ATMega323' 'port.c' 추가



FreeRTOS LED Demo 실습

□ Build Solution



FreeRTOS LED Demo 실습

□ Device Programming

